DAA-1: Fibonacci Series

Techniques: Iteration, Recursion, and Recursion with Memoization

1. Introduction

The Fibonacci series is a sequence in which each number is the sum of the previous two numbers. The series starts with 0 and 1. It is commonly used to understand iteration, recursion, and dynamic programming.

Formula: F(0) = 0, F(1) = 1, and F(n) = F(n-1) + F(n-2) for n ≥ 2.

2. Methods

1. Non-Recursive (Iterative) Method: Uses a loop and two variables to generate each next term. It updates the variables after printing each term.

2. Recursive Method: Calls the same function for the two previous terms. It stops when n is 0 or 1, which are the base cases.

3. Recursive Method with Memoization: Uses recursion along with a dictionary to store calculated Fibonacci values. If a value is already stored, it is reused instead of recalculated.

3. Example with Calculation

Find the first 7 terms of the Fibonacci series.

Calculation:

F(0) = 0
F(1) = 1
F(2) = F(1) + F(0) = 1 + 0 = 1
F(3) = F(2) + F(1) = 1 + 1 = 2
F(4) = F(3) + F(2) = 2 + 1 = 3
F(5) = F(4) + F(3) = 3 + 2 = 5
F(6) = F(5) + F(4) = 5 + 3 = 8

Fibonacci Series: 0, 1, 1, 2, 3, 5, 8

4. Working of the Three Methods

Iterative: Start with a = 0 and b = 1. Print a, then update a = b and b = a + b (using the old values). Repeat for the required number of terms.

Simple Recursion: To calculate F(6), the function calls F(5) + F(4), and those calls continue until the base cases F(1) and F(0) are reached. Some values are calculated repeatedly.

Memoization: When F(2), F(3), and other values are calculated, they are saved in a dictionary. Later calls reuse those stored values, reducing repeated work.

5. Time and Space Complexity

Method	Time Complexity	Space Complexity
Iterative	O(n)	O(1)
Simple Recursion	O(2■)	O(n)
Recursion with Memoization	O(n)	O(n)

6. Applications

• Understanding recursion and dynamic programming
• Algorithm design and complexity analysis

• Mathematical modelling and sequence-based problems

7. Important Viva Questions

Q1. What is the Fibonacci series?
A sequence where each term is the sum of the previous two terms.

Q2. What is the base case in recursion?
F(0) = 0 and F(1) = 1.

Q3. What is memoization?
Storing previously calculated results to avoid repeated calculations.

Q4. Which method is fastest among these three?
The iterative and memoized methods both take O(n) time; iteration uses less auxiliary space.

Q5. Why is simple recursion inefficient?
Because it recalculates the same Fibonacci values many times.

8. Conclusion

The Fibonacci series can be generated using iteration, recursion, and memoization. Memoization improves recursive performance by storing previously computed values, while the iterative method provides linear time with constant auxiliary space.